Project Design Phase
Solution Architecture

Date	08 April 2026
Team ID	SWTID-2026-8559
Project Name	Enforcing Mandatory Fields Using UI Policies and Migra ng Changes with Update Sets
Maximum Marks	4 Marks

Solution Architecture:

The proposed solution architecture is designed to automatically enforce mandatory fields using UI Policies and safely migrate configuration changes using Update Sets in ServiceNow.

This architecture ensures dynamic form validation, reduced manual effort, and error-free deployment across environments.

Architecture Flow

1. User Interaction (Form Input)

- Users (Admin / IT Agent) interact with ServiceNow forms
- Input data is entered during record creation (e.g., Incident form)

2. UI Policy Configuration

- UI Policies are created to define:
  - Mandatory fields
  - Field visibility
  - Read-only conditions
- Conditions are applied dynamically based on user input

3. Dynamic Field Enforcement

- Fields automatically become:
  - Mandatory
  - Visible / Hidden
  - Read-only
- Prevents users from submitting incomplete forms

4. Form Validation Layer

- System validates data in real-time

- Ensures all required fields are filled correctly
- Displays error messages for invalid inputs

5. Update Set Creation

- All configuration changes (UI Policies, form updates) are captured
- Update Sets act as a **version control mechanism**

6. Update Set Tracking & Management

- Changes are tracked with:
  - Name, timestamp, and user details
- Ensures transparency and auditability

7. Migration Across Environments

- Update Sets are moved from:
  - Development → Testing → Production
- Ensures controlled deployment

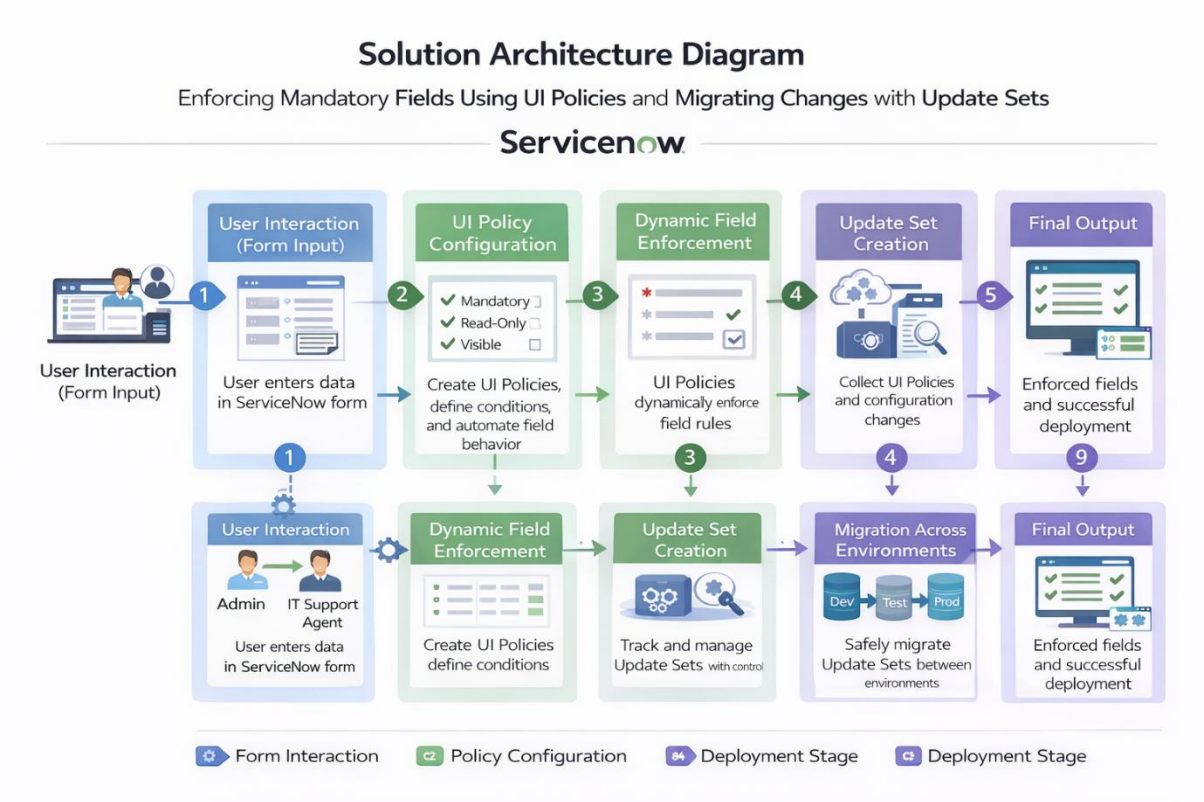
8. Pre-Migration Validation

- System checks for:
  - Conflicts
  - Missing dependencies
- Prevents migration failures

9. Final Output

- Forms enforce mandatory fields automatically
- Configuration changes are successfully migrated
- System ensures:
  - Data consistency
  - Error-free deployment
  - Improved user experience

Example - Solution Architecture Diagram:



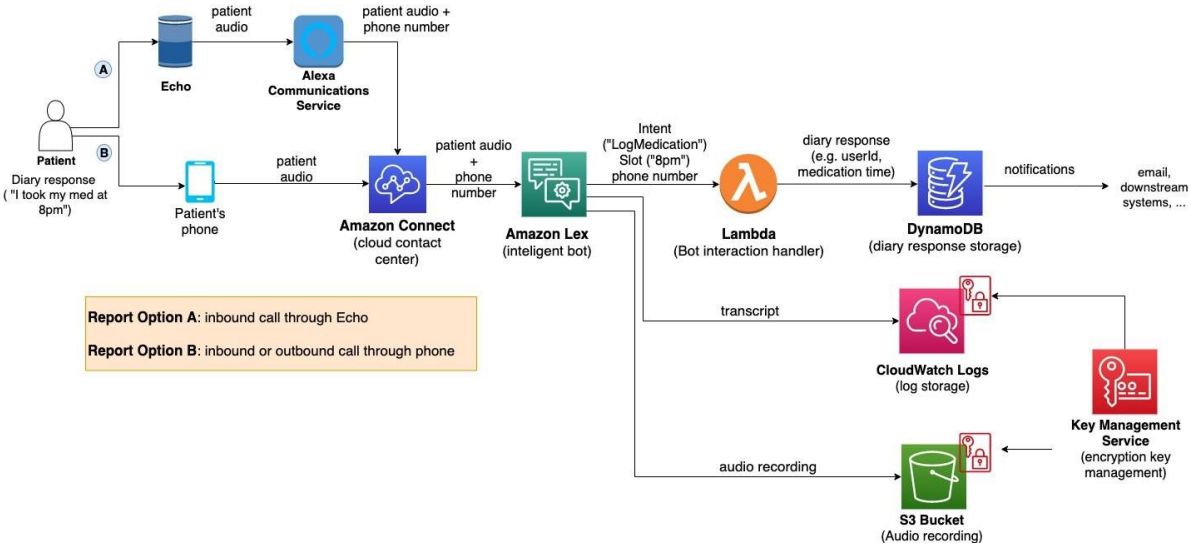


Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/